

Using codes

We use numbers made up of ten numerals (0, 1, 2, 3, 4, 5, 6, 7, 8, and 9), but computers use only two numerals: 0 and 1. This is because computer circuits store data in the form of switches. Each switch holds a single "bit" of information. If the switch is on, this information is a 1; if the switch is off, it is a 0. This means that all information must be coded for the computer as 1s and 0s. This leads to very long numbers, so, to make it easier for humans to handle, binary is often converted into hexadecimal (base 16) numbers.

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

△ Conversion table

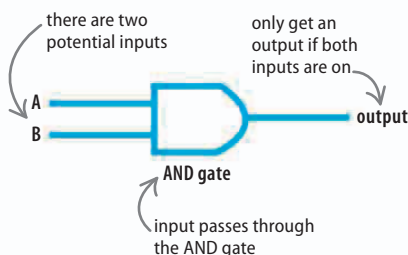
This table shows the decimal (base 10) number system we use converted to binary (base 2) numbers and hexadecimal (base 16) numbers.

The first electronic component was the **diode**, invented in 1904 by English scientist Ambrose Fleming.

Logic gates

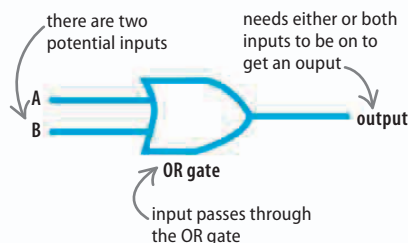
A logic gate is used to make a simple decision. It accepts an electrical signal from its inputs (it can have one or two) and then outputs either an "on," high-voltage signal (representing 1) or an "off," low-voltage signal (representing 0). In computers and many other electronic devices, large numbers of logic gates are linked together to form complex circuits. The image below shows three commonly found logic gates and their possible inputs and outputs.

▷ **AND gate**
There will only be an output from the gate if both inputs are on.



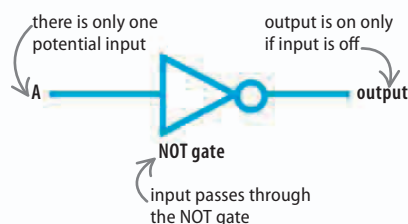
AND gate		
Input A	Input B	Output
1	0	0
0	1	0
0	0	0
1	1	1

▷ **OR gate**
There will be an output if one or both inputs are on.



OR gate		
Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

▷ **NOT gate**
The output is on only if the input is off. If the input is on, the output is off.



NOT gate	
Input	Output
0	1
1	0

REAL WORLD

Retinal implant

Modern electronic devices can be so small, reliable, and sensitive that they can be implanted in the human retina to help some partially sighted people see. Light falling onto the implant is converted into electrical signals that stimulate the optic nerve. The brain interprets these signals as patterns of dark and light, and allows the patient to see objects.

